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Reinforcement Learning for Analog Sizing Optimization

Michel Chevalier*, Severin Trochut*, Roberto Guizzetti*, Pascal Urard*,
Lioua Labrak†, John Samuel‡, Rémy Cellier† and Nacer Abouchi†
*STMicroelectronics, Crolles, France, michel.chevalier01@st.com
†Université de Lyon, CPE Lyon, Institut des Nanotechnologies de Lyon, Lyon, France, lioua.labrak@cpe.fr
‡Université de Lyon, CPE Lyon, LIRIS, Lyon, France, john.samuel@cpe.fr

***Abstract*—This paper proposes a novel adaptation of Machine Learning (ML) techniques based on reinforcement learning for the resolution of analog sizing optimization problems. The paper details the proposed solution and highlights its performances using benchmark tests based on classical analog designs such as two-stages, rail-to-rail and folded-cascode amplifiers. This novel adaptation of the ML technique is used for the optimal design of analog circuits. SPICE simulations are used to verify the viability of the proposed algorithm and novel solutions. The GNN-FCNN approach is validated on a complex circuit with more than 50 devices**

***Index Terms*—analog design, optimization, machine learning, reinforcement learning**

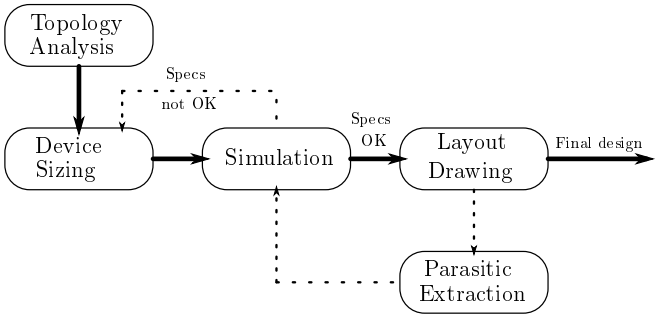


Fig. 1. Overview of the analog pipeline.

I. Introduction

Unlike digital circuit design, analog design demands lots of manual work and expert knowledge, which often makes it the bottleneck of the conception chain [6]. Several reasons make the analog design much harder to work with. First, analog circuits have a larger design space in terms of device size and topology than digital circuits. Second, the specifications of analog design are variable for different applications. Third, analog signals are more susceptible to noise and process-voltage-temperature (PVT) variations, which cost additional efforts in validation and verification. Some tools, like WiCkeD [10], have been developed to address the problem of device sizing, one of the biggest overheads in the design now. The heuristic and iterative nature of WiCkeD makes it a good candidate to apply optimization algorithms such as genetic algorithm [11] or particle swarm optimizers [12]. However, facing the computation time of these solutions, neural networks [1] are beginning to be used to tackle this issue, through their adaptability, and quick computation time after the learning phase. More specifically, genetic algorithms [7], or particle swarms algorithms [8] have been used to tackle the problem. Even though [9] shows good results of optimization with the solutions, they still present an issue of computation time. Then, [5] proposed a method based on neural networks (NN). Even if the training phase of the NN was time consuming, the inference phase was shorter. However, supervised learning does not suit to the ADO process [1] because this method is not able to cover the full complexity of the optimization process. [1] proposed an approach using **Reinforcement Learning** (RL) to face this issue. **RL** methods allow to build adaptative models, that learn how to reach different goals after the learning phase. Even if the results of [1] were better than the previous ones, [2] shows the lack of mathematical representation in [1]. [2] proposed a model of analog circuit based on graphs, and used it to achieve better performances. However, those solutions are still focusing on small electronical topologies (between 10 and 20 components) and have a long learning phase. This paper presents a RL method based on a graph representation of analog circuits, parallelizing the learning algorithm to improve the computation time of the learning phase and supporting more than 50 devices (40 nm CMOS). The next part presents the methodology used and the improvements done in this work. Section III presents an analysis of the obtained results.

II. Methodology

A. ADO Challenge

The aim of Analog Design is to start from a set of specifications (specs) S (Gain, Consumption, Bandwidth, ...), select a schematic and match the specs through the optimization of the devices' parameters, such as the width of a transistor, the value of a capacitor, etc. In this work, the schematic of the amplifier is previously fixed. The aim, therefore, is to optimize the devices' parameters to reach

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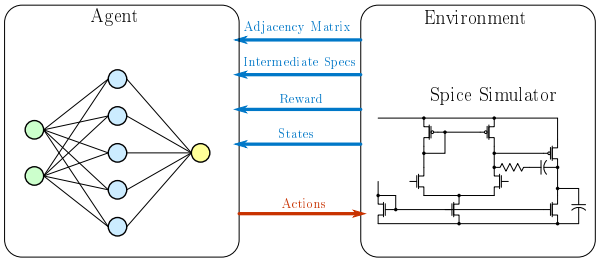


Fig. 2. RL Agent-Environment.

the targeted specs. The mathematical model of this study is defined as follows:

$$arg_x min \sum_x q_c(x), s.t. f(x) \geq y \quad (1)$$

Where $x \in \mathbb{R}^n$ defines the design parameters, where n represents the number of features that can represent an electronical component, including the size of each transistor, the capacitor value, and the resistor. $y \in \mathbb{R}^m$ are the target specifications, where m stands for the number of electrical specifications, such as the DC gain, the bandwidth, etc. The simulator f is defined as the map from parameters to the specifications. $q_c(x)$ is the objective function. While Analog Designers work through a trial-and-error process to optimize an amplifier, the Reinforcement Learning (RL) [13] seems to be well suited for this study.

B. Reinforcement Learning

In a typical RL framework, two main entities interact: an agent and an environment [1] (Fig 2). The agent aims to optimize a reward, given by the environment, through this interaction by taking actions. The agent gathers information from the environment to reach a target. The aim is for the agent to be able to perceive the characteristics of the environment, and can perform actions to modify its state. The quality of the action taken by the agent according to the current state is rated by the reward. The better the action was according to the current state of the agent, the higher is the reward. In this case, a neural network has been used as the agent, and the Eldo SPICE simulator as the environment.

C. Environment

The environment is the part of the system that updates the properties of the circuit components based on some actions provided by an agent. Naturally, the new properties will change the intrinsic behavior of the circuit. To measure the effect of this new configuration, the environment uses a simulation framework, which in this case is a SPICE-like simulator. This simulator outputs the new specification values, which are compared with targeted values through a custom function to compute the reward value. This value and the last specifications are sent to the agent to be further processed. An overview of the environment and its components is shown in Fig 3.

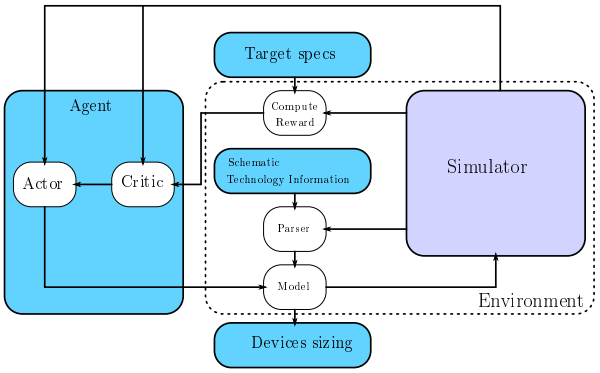


Fig. 3. Use of Actor - Critic structure in the proposed approach.

D. Agent

For the agent, a neural network has been chosen with an Actor-Critic structure [2] Fig 3. Actor-Critic are made of two neural networks interacting with each other. The first one is taking actions according to the environment's information and the current state, and the second one is giving an estimation of the value function along the agent's path. The Actor and the Critic networks have a similar structure Fig 4. They are composed of a dense neural network leveraging the general information of the circuits (gain, consumption, bandwidth,...) of 3 layers (128-64-64), a graph convolutional NN [3] of 7 layers, to embed the topology information of the circuit, and another dense neural network (128-64- output layer) giving the actions distribution or the estimated value function.

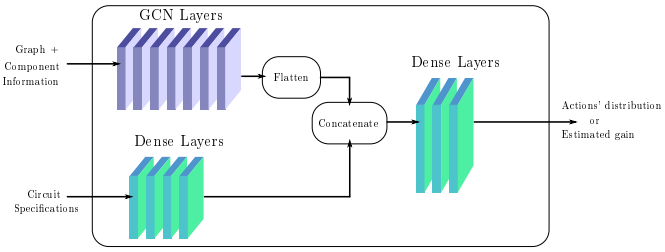


Fig. 4. Detailed view of Actor & Critic Structure.

E. Graph Convolutional Neural Network

As our Actor-Critic agent works with integrated analog circuits, it uses a useful mathematical representation, a graph [2]. This graph is composed of nodes and edges (in the same way a circuit is made of components and connections). Each node has several so-called features, which in our case are representative parameters of our transistors, the width, the length and the multiplicity, or other components (resistors or capacitors). A general node is then characterized by its features and the number of connections (e.g., number of direct neighbors). Fig 5 shows the schematic of a two-stage operational amplifier. After extracting the features of each component, it is

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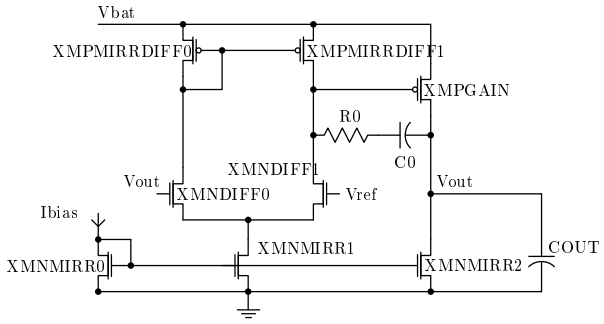


Fig. 5. Two-stage operational amplifier.

then transformed into the graph shown in Fig 6. To

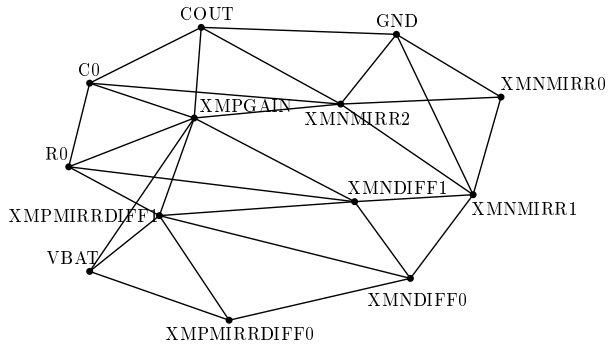


Fig. 6. Graph of the Two-stage operational amplifier.

extract the topological information from this graph, graph convolutional neural network [3] is an efficient tool. The convolution defined below allows to gather the features from the neighbors of the component to its own features.

$$H^{l+1} = relu(D^{-1/2}AD^{-1/2}H^{l}W^{l+1}) \quad (2)$$

Where H^{l+1} is the output $l+1th$ layer, D is the matrix of degree of the graph, A is the adjacency matrix, and W^{l+1} the weights of the $l+1th$ layer. Through this aggregation of features, the current node has a better description of its surrounding, allowing a better decision for the actions taken in this case [2].

F. Training

The training stage is an iterative process where the weights of all the networks in the agent modules are tweaked so that the Actor provides good actions that lead to desired specifications. Some of the most successful RL algorithms available today belong to the Policy Gradient family. Specifically, the Proximal Policy Optimization (PPO) [1] has been used in this work. This method uses multiple epochs of stochastic gradient ascents to perform explicit policy updates at each step. To build the dataset to adjust the weights of the neural network, several instances of the Spice simulator has been launched. A set of gradient ascents applied from the same dataset to the agent is called an iteration. The training is composed of several iterations, in order to update the agent with different datasets.

III. Experiments and Results

A. State-of-the-Art Experiment

To compare GNN-FCNN (Graph NN - Fully Connected NN) with state-of-the art results, the experiment from [4] has been chosen. It a two-stage simple operational amplifier (Simp. Op. Amp.) Fig 5 as benchmark, involving 12 components. The training period for the RL agent is 3.5 10^4 iterations, the maximum episode length has been set to 50 steps. To produce the dataset used by the PPO algorithm, 16 CPUs have been used to run 16 instances of the SPICE-like simulator ELDO, to mimic the 16 RL agents in parallel. To test the efficiency of the outputted RL agent, a dataset of 200 sets of specifications was randomly sampled in the ranges described in Tab I. As a starting point, all the transistors are set to be in the right mode.

TABLE I
[4] Experiment Context

Circuit	Two-stage Op-Amp			
Technology	40 nm CMOS			
# of features	$2 \times 7 + 1 = 15$			
Design space	W (μm) [1, 100]	# of fingers [2, 32]	C (pF) [0.1, 10]	
Specs	Gain [300, 500]	Bandwidth $[10^6, 2.510^7]$ Hz	Phase margin [55o, 60o]	Power consumption $[10^{-4}, 10^{-2}]$ W

B. Results

According to Tab II, GNN-FCNN is faster in term of training time, and is able to achieve the same performances as shown in [4]. However, this experiment focuses on a simple schematic, with a reduced set of tunable parameters.

TABLE II
Experiments comparison

Method	Training Time	Efficiency on test
[4]	5h	99%
Our work	**20 min.**	99%

C. Full Complexity Experiment

While the first experiment showed promising results, a second one has been designed to implement deeper test of this method. In the first experiment, some components' parameters were fixed from the starting point, while the aim of this second experiment is to show that GNN-FCNN is able to size an amplifier in its full complexity. To do so,

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two circuits have been selected to run the experiment: the Simple Op Amp Fig 5, and a rail-to-rail amplifier with AB class output buffer Fig 7.

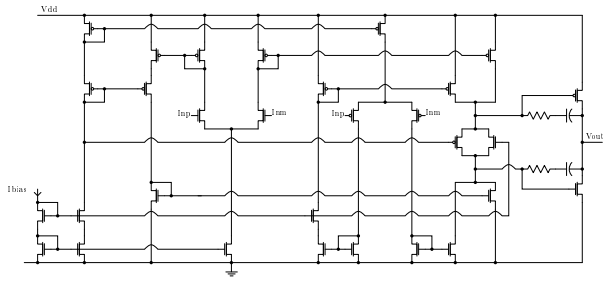


Fig. 7. Rail-to-rail input opamp.

In this experiment, the starting point is set as a working point for the amplifier. The training period for the RL agent is 500 iterations, the maximum episode length has been set to 250 (170 for the rail-to-rail amplifier) steps. To test the efficiency of the outputted RL agent, a dataset of 100 sets of specifications was randomly sampled.

D. Results

TABLE III
FULL COMPLEXITY EXPERIMENT RESULTS

Design type	Simple Amplifier 10 devices 19 spec points	Rail-to-rail Amplifier 56 devices 38 spec point
Learning time for AI	4h	7h
Inference time	27 sec.	46 sec.
Existing EDA solution [10]	24 min.	1H10
Estimation for a senior designer	1 or 2 days	20 days
% of solutions achieving the specification	87%	100%

Tab III shows GNN-FCNN is able to solve the optimization problem in its full complexity, such as WiCkeD. Even if these tool looks faster compared to the training time of GNN-FCNN, the inference shows that optimizing an amplifier would be way faster. It is also able to size more complex designs, compared to other NN methods.

IV. DISCUSSION

According to the results shown in Tab II and Tab III, this method is able to optimize efficiently Op Amp. Regarding to other methods, it appears that GNN-FCNN is able to size complex architectures, such as the rail-to-rail amplifier. However, the stability of the learning phase has to be improved, it may be done by a better representation of the components' features. Furthermore, EDA tools can be faster compared to the learning time of this solution, and may be more attractive to size small topologies. But according to the iterative process of ADO, GNN-FCNN showed promising results on complex architectures with its inference time.

V. CONCLUSION

In this paper, an optimization technique for the optimal sizing of analog circuits based on reinforcement learning has been proposed. The corresponding methodology was validated by various tests and applied to optimize performances of analog circuits, namely several variants of CMOS amplifiers. Optimal parameters of the neural network were determined using a statistical approach. Reached performances were validated via SPICE simulations with Siemens ELDO. It was also shown that the proposed methodology gives better results in terms of computing time and optimum quality, when compared to previous works.

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